



OUR WORLD IN DATA

Global Renewable Energy in 2025

How Far Have We Come?

A data-driven overview of renewable energy's growing share in the global energy and electricity mix

Sources: Ember (2026); Energy Institute – Statistical Review of World Energy (2025)

For policy analysts and energy sector strategists

75% of Global GHG Emissions Still Come from Fossil Fuels



75%

of global greenhouse gas emissions come from burning fossil fuels for energy



5M+

premature deaths per year from air pollution caused by fossil fuel combustion

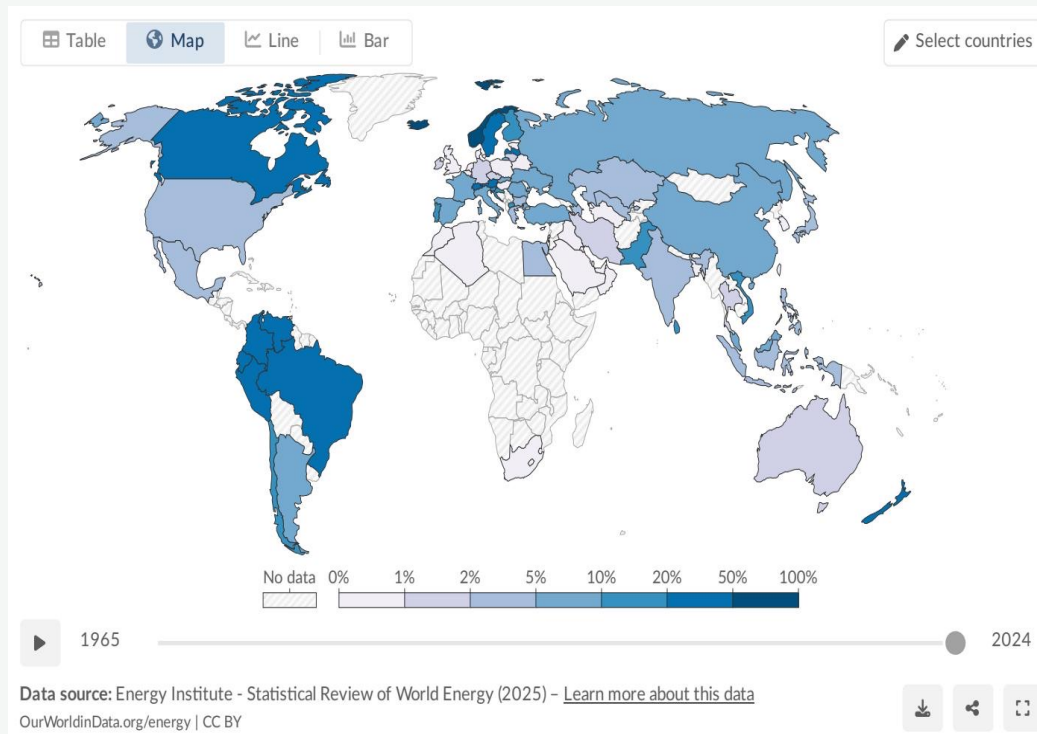
Since the Industrial Revolution, fossil fuels have dominated the world's energy systems. To reduce CO₂ emissions and air pollution, the world must rapidly shift toward low-carbon energy sources — nuclear and renewable technologies.

Renewables Now Supply ~14% of Global Primary Energy

~14%

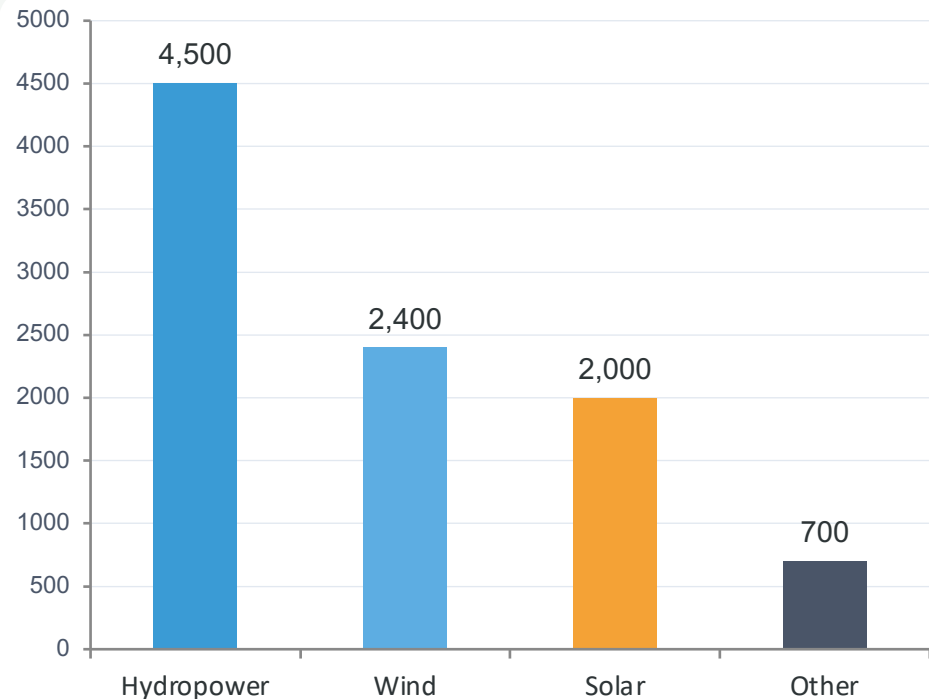
of the world's primary energy comes from renewable technologies (2024, substitution method)

*Primary energy = electricity + transport + heating.
Transport and heating remain heavily reliant on oil and gas, keeping the renewable share lower than in electricity alone.*



Share of primary energy from renewables by country, 2024

Renewable Electricity Generation Is Growing Rapidly



Technology	TWh	Share
Hydropower	~4,500	47%
Wind	~2,400	25%
Solar	~2,000	21%
Other	~700	7%

Wind + Solar combined (~4,400 TWh) now nearly match Hydropower's output

Almost One-Third of Global Electricity Now Comes from Renewables

Share of electricity production from renewables, 2025

Renewables include solar, wind, hydropower, bioenergy, geothermal, wave, and tidal sources.

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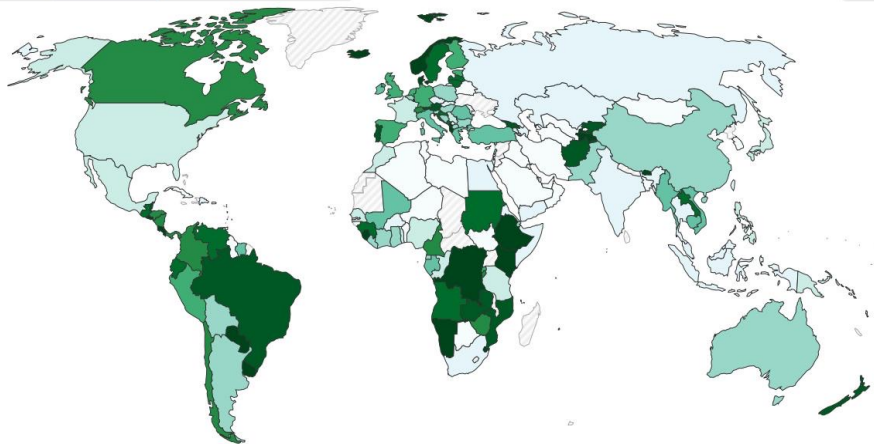
Table

Map

Line

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Select countries



No data 0% 10% 20% 30% 40% 50% 60% 70% 80% 90% 100%



1985



2025

Data source: Ember (2026); Energy Institute - Statistical Review of World Energy (2025) - [Learn more about this data](#)

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Country	RE Share
Norway	~98%
Brazil	~85%
Canada	~65%
China	~35%
India	~25%
Saudi Arabia	<2%
World avg.	~30%

Share of electricity from renewables by country, 2025

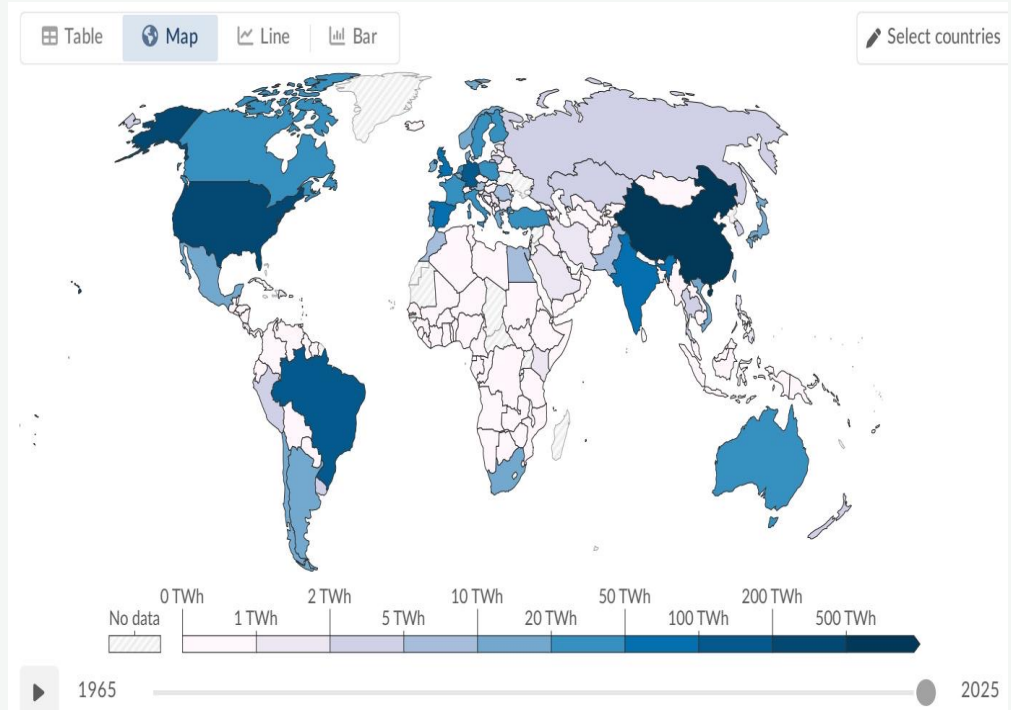
Hydropower Remains the Largest Renewable Source at ~50% of Generation

~4,500 TWh

Hydropower has been generating electricity at scale for over a century and accounts for roughly half of all renewable generation.

Top producers:

China • Brazil • Canada



Hydropower generation by country, 2025

Wind and Solar Are the Fastest-Growing Energy Technologies in History



Wind Energy

~2,400 TWh | 25% of renewable electricity

Major contributor in Europe, China, and the US.
Includes onshore and offshore.

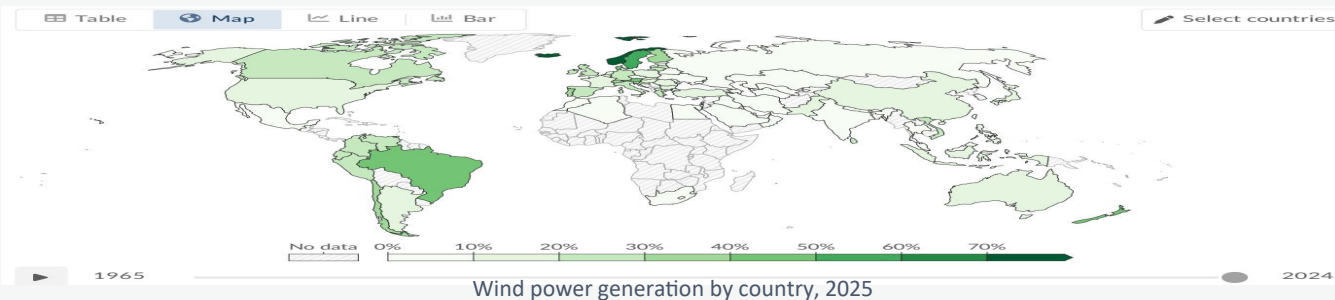


Solar Energy

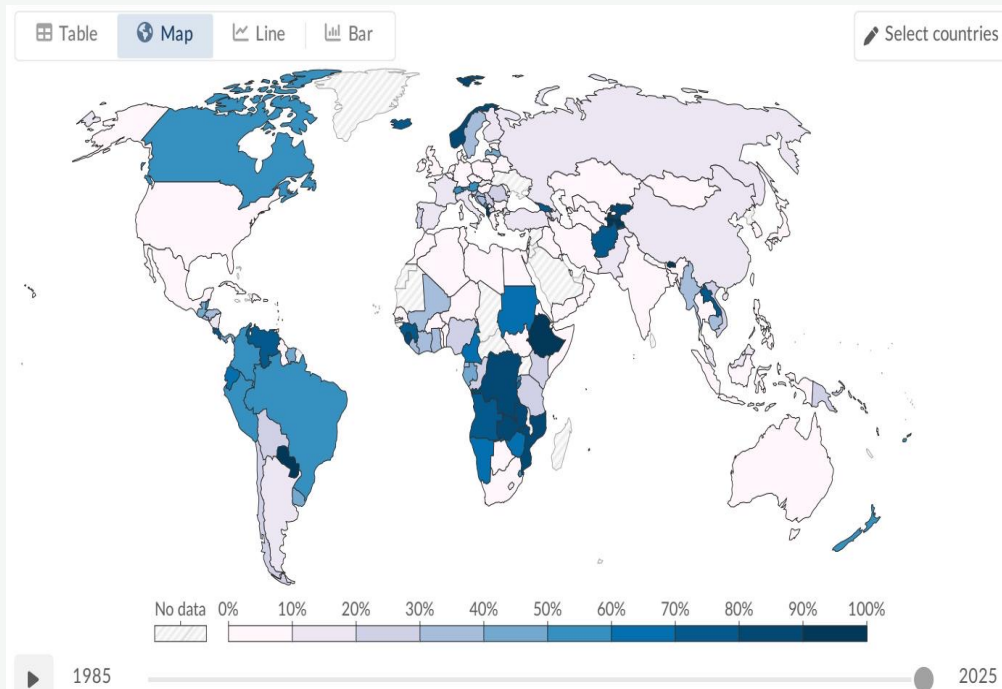
~2,000 TWh | 21% of renewable electricity

Fastest-growing source. Dramatic growth since 2010,
driven by falling costs and policy support.

Combined wind + solar: ~4,400 TWh — nearly matching hydropower's ~4,500 TWh



Stark Regional Gaps in the Energy Transition



Share of electricity from hydropower by country, 2025

Leaders (60–98% RE):

Scandinavia, Brazil, Canada benefit from abundant hydropower; Denmark & Germany invested heavily in wind/solar

Lagging (<10% RE):

Sub-Saharan Africa has minimal renewable infrastructure; Middle Eastern nations remain almost entirely fossil-fuel dependent

The transition to renewables is highly uneven across the globe.



Key Takeaways

Renewables Are Growing Fast but the Transition Must Accelerate

- ✓ ~**14% of primary energy** and ~30% of electricity now come from renewable sources — significant but far from climate targets.
- ✓ **Hydropower remains dominant** at ~50% of renewable generation, but its growth potential is limited vs. wind and solar.
- ✓ **Wind & solar are the fastest-growing** energy technologies in history; combined output nearly matches hydro.
- ✓ **Stark regional disparities persist:** Scandinavia/Latin America at 60–90%+ vs. Middle East/Sub-Saharan Africa below 10%.
- ✓ **Transport & heating lag far behind** electricity, keeping overall primary energy share much lower.

The pace of the renewable transition must accelerate to meet global climate targets. Every policy decision matters.